INCH-POUND
MIL-V-868/2(SH)
21 July 1989

MILITARY SPECIFICATION SHEET

VALVES, HYDRAULIC DIRECTIONAL CONTROL, FOUR-WAY, 3 - 25 GPM

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-V-868.

1. GENERAL

1.1 Classification. Directional control valves covered by this specification sheet shall be designated using Part or Identifying Numbers (PIN) as follows (see 1.1.8 for examples of PINs):

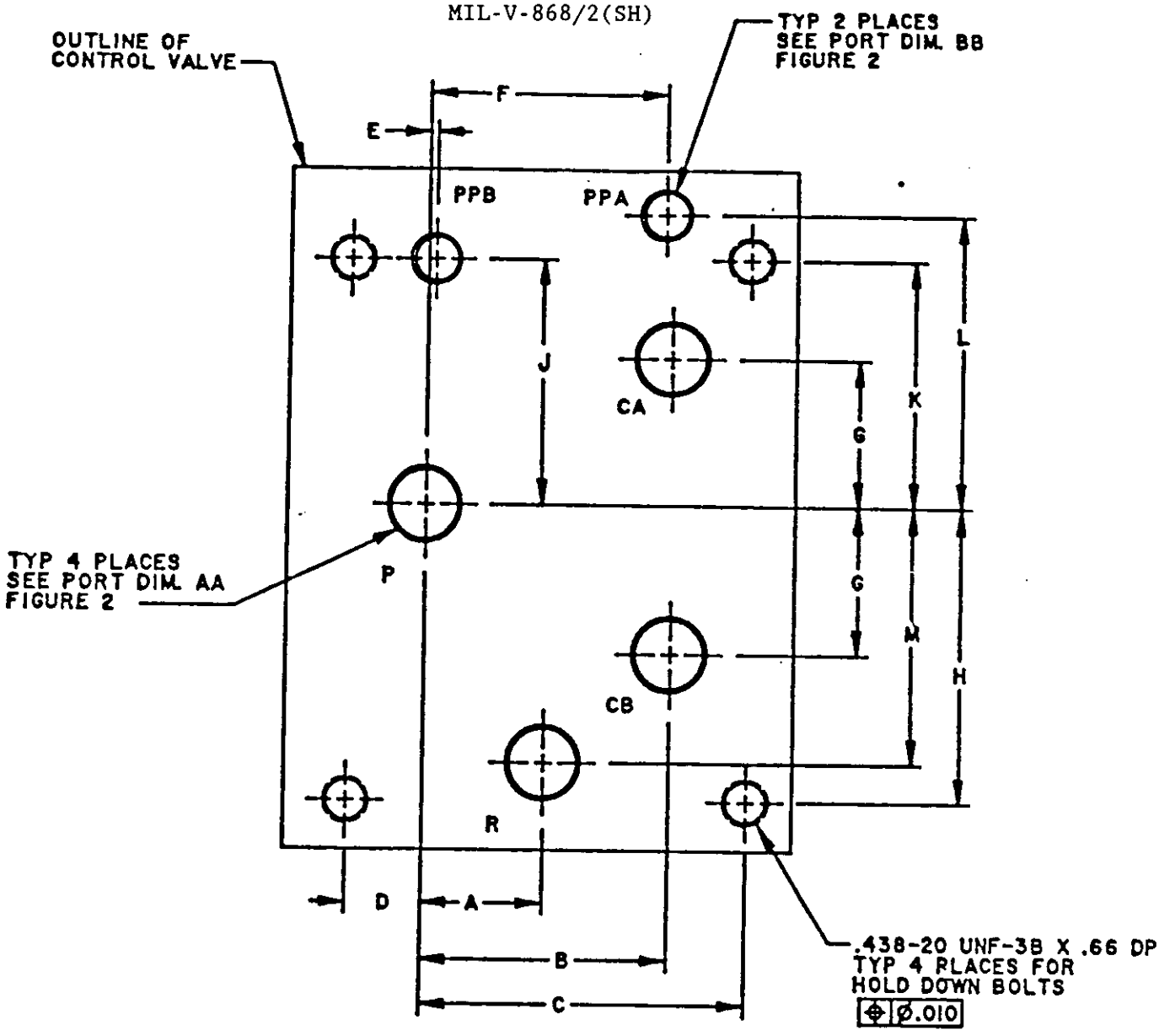
M	868	/2	ZZ	T	YYY	ZZ	X
Prefix to indicate a Military specification	Specification number	Applicable specification sheet number (see 1.1.1)	Left side operator (see 1.1.2)	Spool type (see 1.1.3)	Spool flow paths (see 1.1.4)	Right side operator (see 1.1.5)	Noise and fatigue (see 1.1.6)

1.1.1 Applicable specification sheet number. The applicable specification sheet number designates the valve's performance characteristics (flow range, pressure losses, leakage, etc.) and mounting interface configuration. Figures 1, 2, and 3 define the mounting interface configuration and maximum dimensions for valves with various types of operators. Control valves covered by this specification sheet shall be of the integral slide and sleeve type.

1.1.2 Left side valve operator. The left side valve operator shall be designated by a two character code. The left hand operator shall be designated as the actuator shown to the left of the spool symbol. On the actual valve, this corresponds to the actuator closest to the return port. For valves operated by a handle or lever, the two character code shall be in accordance with table I. The first character designates the type of operator in addition to the handle. The second character is a number (0 to 7) and indicates the positions, if any, in which the valve spool may be latched or detented. For valves without an operating handle, the left side operator shall be designated by a two character code in accordance with table II. The first letter designates the primary operator and the second letter identifies secondary operators, if any. See 1.1.7 for special operator configurations not covered by tables I and II.

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P = PRESSURE PORT
R = RETURN PORT
CA = CYLINDER PORT
CB = CYLINDER PORT
PPA = PILOT PRESSURE PORT A
PPB = PILOT PRESSURE PORT B

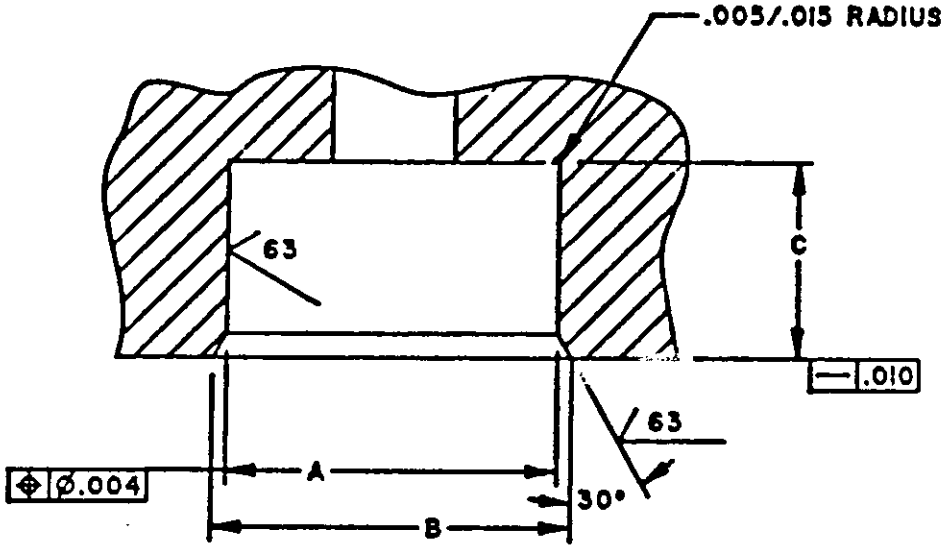
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DIMENSION IN INCHES					
A	B	C	D	E	F
1.000	2.000	2.625	.625	.062	1.938
G	H	J	K	L	M
1.188	2.375	2.000	2.000	2.375	2.063

SH 13232033

FIGURE 1. Valve mounting configuration.

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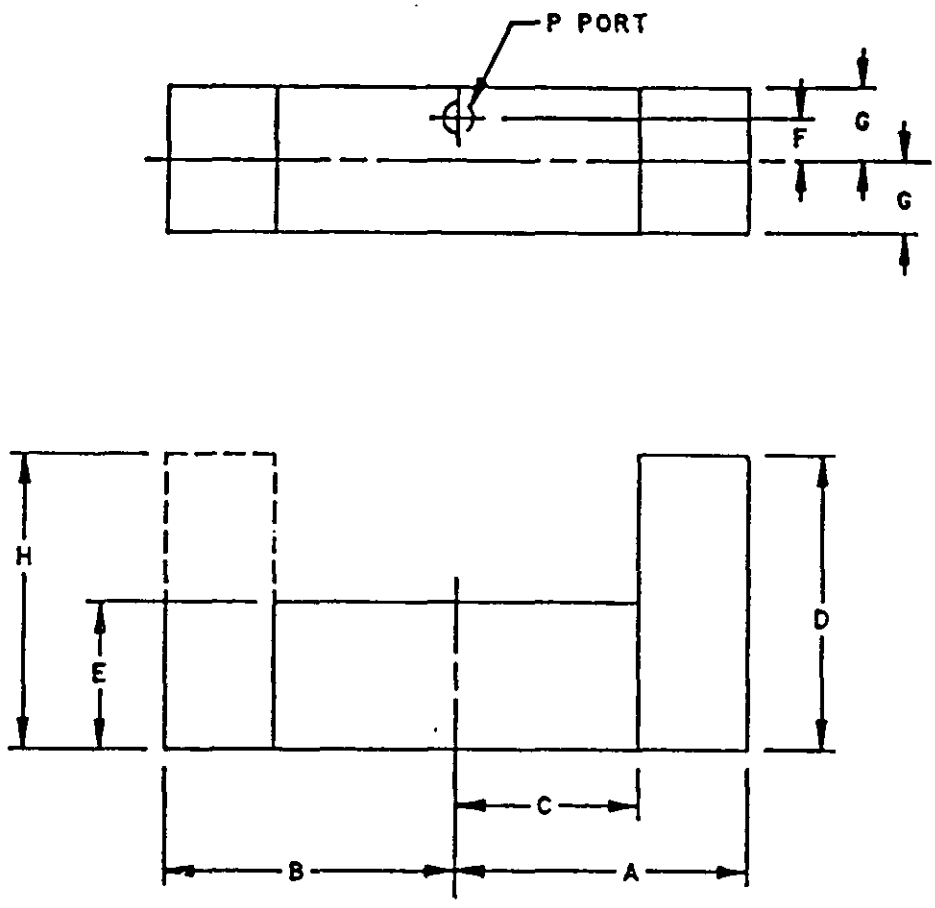


COUNTERBORE DIMENSION IN INCHES			
PORT	A	B	C
AA	.991/.993	1.030/1.060	.440/.446
BB	.360/.361	.390/.420	.338/.344

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FIGURE 2. Counterbore dimension.

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DIMENSIONS IN INCHES

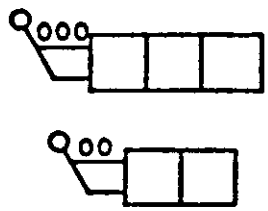
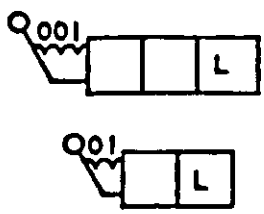
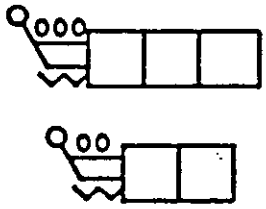
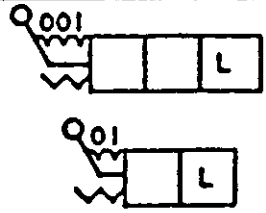
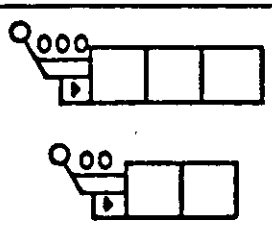
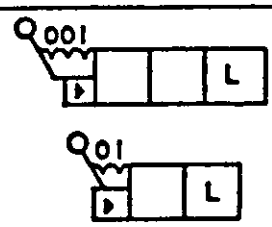
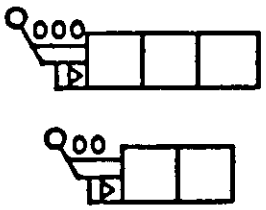
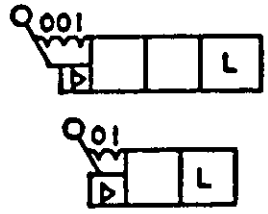
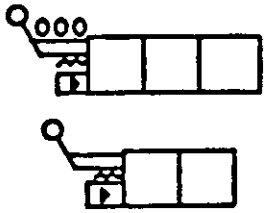
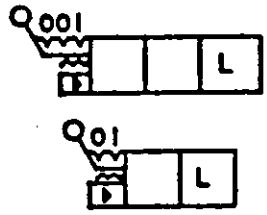
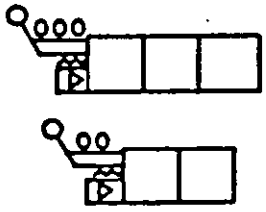
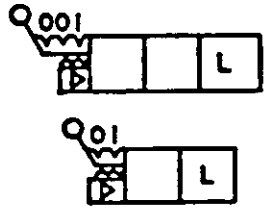
	A MAX	B MAX	C MIN	D MAX	E MAX	F MAX	G MAX	H
DOUBLE SOLENOID	8.63	8.63	3.20	7.50	5.00	1.02	2.25	7.50
SINGLE SOLENOID	8.63	8.63	3.20	7.50	5.00	1.02	2.25	-
SOLENOID OR HYD PILOT WITH LEVER OVERRIDE	12.63	9.88	3.20	9.50	5.00	1.02	2.25	7.50
LEVER OPERATION	11.00	10.00	3.20	9.50	5.00	1.02	2.25	-
MECHANICAL	8.70	8.70	3.20	5.00	5.00	1.02	2.25	-

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FIGURE 3. Valve envelope.

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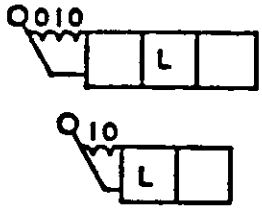
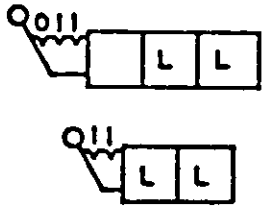
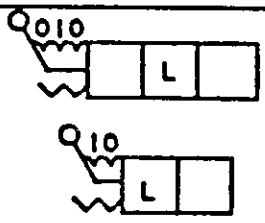
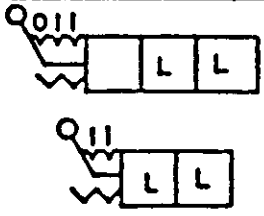
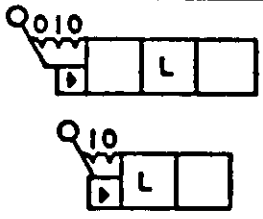
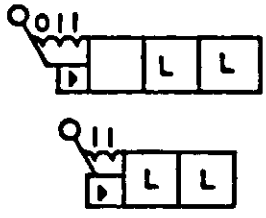
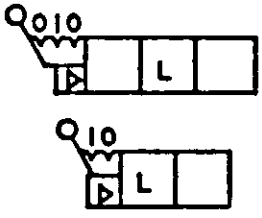
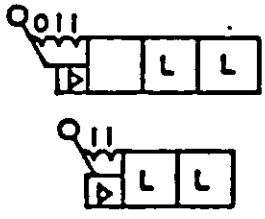
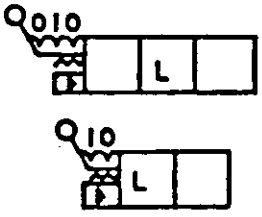
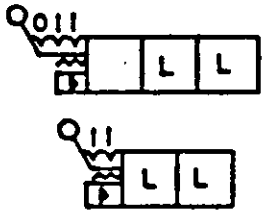
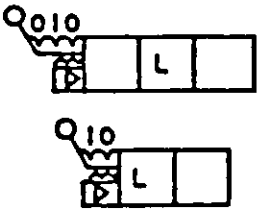
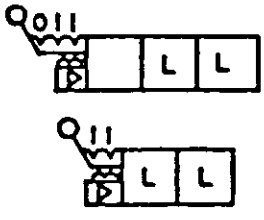
TABLE I. Valve operators-lever.

FIRST CHARACTER	SECOND CHARACTER 0	SECOND CHARACTER 1
LEVER ONLY L		
SPRING RETURN S		
HYD. PILOT H		
PNEU. PILOT P		
SPRING RETURN HYD. PILOT R		
SPRING RETURN PNEU. PILOT Q		

SYMBOLS "1" AND "L" INDICATES THE POSITION THAT THE VALVE SPOOL IS LATCHED OR DETENTED.

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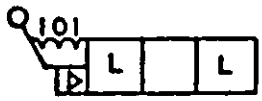
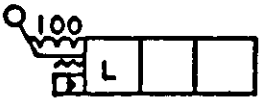
TABLE I. Valve operators-lever - Continued.

FIRST CHARACTER	SECOND CHARACTER 2	3
LEVER ONLY L		
SPRING RETURN S		
HYD. PILOT H		
PNEU. PILOT P		
SPRING RETURN HYD. PILOT R		
SPRING RETURN PNEU. PILOT Q		

SYMBOLS "I" AND "L" INDICATES THE POSITION THAT THE VALVE SPOOL IS LATCHED OR DETENTED.

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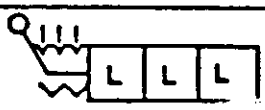
TABLE I. Valve operators-lever - Continued.

FIRST CHARACTER	SECOND CHARACTER 4	5
LEVER ONLY L		
SPRING RETURN S		
HYD. PILOT H		
PNEU. PILOT P		
SPRING RETURN HYD. PILOT R		
SPRING RETURN PNEU. PILOT Q		

SYMBOLS "I" AND "L" INDICATES THE POSITION THAT THE VALVE IS LATCHED OR DETENTED.

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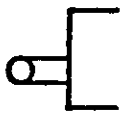
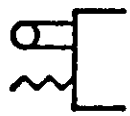
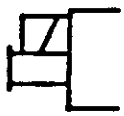
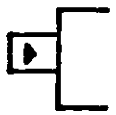
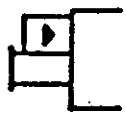
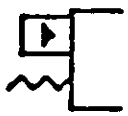
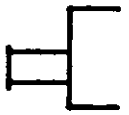
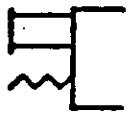
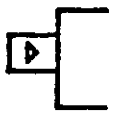
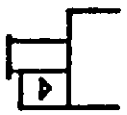
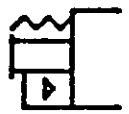
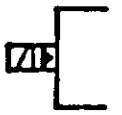
TABLE I. Valve operators-lever - Continued.

FIRST CHARACTER	SECOND CHARACTER 6	7
LEVER ONLY L		
SPRING RETURN S		
HYD. PILOT H		
PNEU. PILOT P		
SPRING RETURN HYD. PILOT R		
SPRING RETURN PNEU. PILOT Q		

SYMBOLS "I" AND "L" INDICATES THE POSITION THAT THE VALVE SPOOL IS LATCHED OR DETENTED.

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TABLE II. Valve operators.

FIRST LETTER	SECOND LETTER (ADDITIONAL OPERATORS)			
	X (NONE)	M (MANUAL)	S (SPRING)	T (SPRG RET MAN OVRRID)
CAM C		NA		NA
ELECTRICAL E				
HYDRAULIC H				
MANUAL M		NA		NA
PNEUMATIC P				
SPRING S		NA	NA	NA
TWO-STAGE T				
WITHOUT OPERATOR X		NA	NA	NA

NOTE: ONLY LISTED COMBINATIONS MAY BE USED
NA= NOT AVAILABLE

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Example: ES designates a solenoid operated valve with return spring.
H2 designates a manually operated valve with a hydraulic pilot override which is latched or detented in the center spool position.

1.1.3 Spool metering/adjustment features. The spool metering/adjustment features are designated by a single character as follows:

- Standard valve with no metering or adjustable flow limiters.

M Metering valve where flow is proportional to spool displacement. The maximum fluid flow can be limited by adjustable spool stroke limiters (see 2.1.1 and 2.1.2.4).

A Adjustable valve where fluid flow can be limited by spool stroke limiters (see 2.1.1).

1.1.4 Spool flow paths. Each valve position flow path shall be designated by a number (or character) in accordance with table III. A two- position valve is designated by two numbers (or characters) and a three- position valve is designated by three numbers (or characters). Use of more than three characters to designate spool flow paths shall require approval from the Naval Sea Systems Command. See 1.1.8 for examples of part numbers.

1.1.5 Right side valve operator. The right side valve operator shall be designated by a two character code in accordance with table II. The first letter designates the primary operator and the second letter identifies secondary operators, if any. See 1.1.7 for special operator configurations not covered by tables I and II.

Example: EM designates a solenoid operated valve with manual override.
SX designates a return spring only.

1.1.6 Noise and fatigue grade. The noise requirements and the fatigue requirements of the valve shall be designated by a single letter.

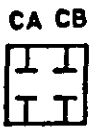
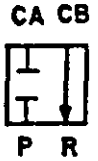
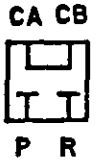
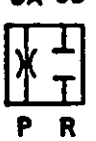
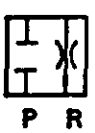
Fatigue rated valves shall conform to the fatigue rated test requirements of MIL-V-868. The single letter designates both the fatigue requirements and the noise grade (see 1.5) of the valve as indicated below:

Non-fatigue rated designation	Fatigue rated designation	MIL-STD-2193 noise level
A	F	A
B	G	B (no noise requirements)
C	Y	C
D	Z	D

The part number assigned a valve shall reflect the most stringent noise level which the valve meets. Accordingly, a noise level D valve would also be suitable in applications requiring noise level A, B, and C valves. Similarly, a fatigue rated valve is suitable for use in a non-fatigue rated application with equivalent or less stringent noise requirements.

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TABLE III. Spool flow path.

CODE	SYMBOL	CODE	SYMBOL	CODE	SYMBOL
0		1		2	
3		4		5	
6		7		8	
9		Z		K	
J					

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1.1.7 Special configuration. Where the required valve configuration cannot be described by the standard designators in tables I, II, and III, the part number designators will be assigned by the Naval Sea System Command following approval of the desired configuration.

1.1.8 Examples of part numbers. Examples of valve part numbers are shown in table IV.

1.2 Mounting configuration. The mounting configuration of valves shall be as shown on figure 1.

1.2.1 Fluid connection. Unless otherwise designated, the quill or sealing sleeve configuration shown on figure 2 shall be used.

1.2.2 Pilot ports. The pilot ports PPA and PPB, shown on figure 1, are used with valves that require external hydraulic pilot supply to shift the spool. These are valves that have operators which are identified by the letter or character codes H, P, R, and Q from tables I and II.

1.3 Envelope dimensions. The valve assembly shall not exceed the limitations of the envelope dimensions listed on figure 3.

1.4 Solenoids. The solenoids shall be of the direct current type with a rectifier to convert incoming alternating current to direct current.

1.5 Noise grade. Valves shall meet the applicable structureborne noise levels as identified in MIL-STD-2193 as designated by the part number (see 1.1.6). A valve shall carry the part number indicating the most stringent noise level which it meets.

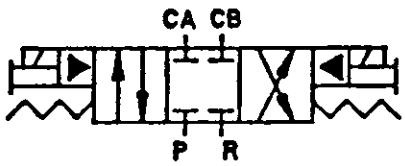
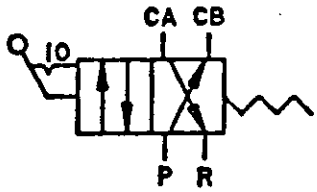
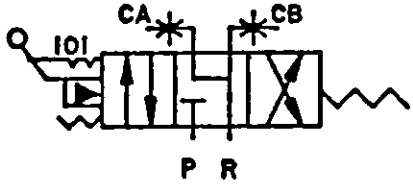
1.6 Hydraulic fluid. The valve shall meet the fluid compatibility requirements specified in MIL-STD-2193 including the immersion test for nonmetallic materials. Fluid in accordance with MIL-L-17331 may be used in lieu of 2135 TH fluid in accordance with MIL-H-17672 for the immersion test.

1.7 Test fluid and temperature. The test fluid shall be either of the two fluids specified in 1.6. Unless otherwise specified for the specific test, the fluid temperature shall be controlled to meet the viscosity requirements identified below. (See MIL-V-868 for guidance regarding fluid viscosity at various test temperatures.)

(a) For leakage test, the maximum fluid viscosity at atmospheric pressure shall be 75 centistokes.
(b) For pressure drop tests, hydraulic actuation test, and solenoid actuation tests, the fluid viscosity at atmospheric pressure shall be between 75 and 97 centistokes.
(c) For noise testing, the fluid viscosity at atmospheric pressure shall not exceed 50 centistokes.

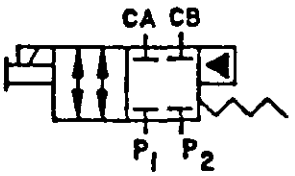
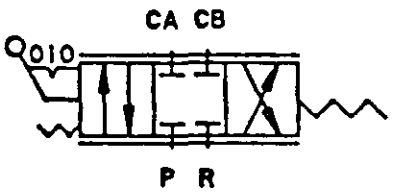
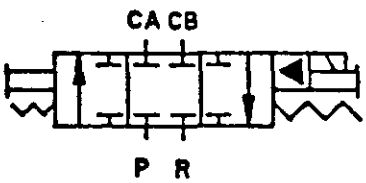
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TABLE IV. Examples of part numbers.

4-WAY, 3-POSITION, DOUBLE SOLENOID, SPRING CENTERED VALVE, WITH CLOSED CENTER SPOOL, MANUAL OVERRIDE. QUIET VALVE OPERATION.    M868/2TT-109TTA
4-WAY, 2-POSITION, MANUAL OPERATED, LATCH IN P TO CA POSITION, SPRING RETURN TO P TO CB POSITION. NON-QUIET VALVE OPERATION.    M868/2L2-19SXB
4-WAY, 3-POSITION, LEVER OPERATED, SPRING CENTERED VALVE FOR NON-QUIET OPERATION. VALVE TO BE EQUIPPED WITH STROKE LIMITERS AND HYDRAULIC OVERRIDE. SPOOL LATCHES IN P TO CA POSITION AND P TO CB POSITION.    M868/2R5A139SXB

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TABLE IV. Examples of part numbers - Continued.

4-WAY, 2-POSITION, SOLENOID OPERATED, QUIET VALVE. MANUAL OVERRIDE TO THE OPEN POSITION, SPRING RETURN TO BLOCKED POSITION. HYDRAULIC PILOT OVERRIDE TO BLOCKED POSITION. (P2 MUST BE LESS THAN OR EQUAL TO 600 PSI.)    M868/2ET-10HSA
4-WAY, 3-POSITION, LEVER OPERATED, SPRING CENTERED, QUIET METERING VALVE. SPOOL LATCHED IN CENTER POSITION. FLOW IS PROPORTIONAL TO SPOOL STROKE.    M868/2S2M109SXA
4-WAY, 3-POSITION, SPRING CENTERED, NON-QUIET VALVE. SOLENOID OPERATED TO DIRECT CB TO R, MANUALLY OPERATED TO DIRECT P TO CA.    M868/2MS-206TTB

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2. PERFORMANCE

2.1 Flow rate. Valves covered by this specification sheet shall have the following range:

(a) Noise grade A, C, & D .... 3 to 12 gallons per minute
(b) Noise grade B ..... 5 to 25 gallons per minute

2.1.1 Flow control adjustment. Valves with flow control adjustments shall be capable of restricting flow to the minimum valves in 2.1, at a minimum differential pressure of 2200 lb/in^2 gauge and a viscosity of 50 centistokes maximum at atmospheric pressure.

2.1.2 Differential pressure at rated flow. The differential pressure drop across a valve with the designated spool style at the maximum rated flow listed in 2.1 shall be as follows.

2.1.2.1 For valves that utilize flow paths 1 and 9 from table III, the differential pressure across the valve in that position shall not exceed 500 lb/in^2 gauge. The pressure differential shall be measured from the supply port (P) to the return port (R) with the cylinder ports (CA and CB) connected by a piping loop.

2.1.2.2 For valves that utilize flow paths 2, 3, 5, 6, and Z from table III, the differential pressure across the valve in that position shall not exceed 250 lb/in^2 gauge. The pressure differential shall be measured from either the supply port (P) or return port (R) to one of the cylinder ports, either CA or CB. The remaining port is blocked.

2.1.2.3 For valves that utilize flow paths 4, 7, and 8 from table III, the differential pressure across the valve in that position shall not exceed 250 lb/in^2 gauge. The pressure differential shall be measured from one of the ports to the other.

2.1.2.4 For metering valves (spool type M), the valve flow characteristics versus spool displacement shall be within the requirements as specified in table V.

2.2 Solenoid actuation. The solenoid actuation rate specified in MIL-V-868 shall be met with the valve at ambient room temperature under the following conditions:

Condition	Supply pressure lb/in^2 gauge	Return pressure lb/in^2 gauge	Shift time milliseconds
1	2800 minimum	100 maximum	250 minimum
2	1200 maximum	100 minimum	500 maximum

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TABLE V. Metering valve flow.

Percent spool stroke	Percent rated flow at 500 lb/in^2 differential	
	maximum	minimum
5	5	0
10	10	2
20	20	5
40	40	16
60	60	32
80	--	50
100	--	100

2.3 Leakage. Leakage rate shall not exceed values listed in table VI.

2.4 Noise. The structureborne noise for all valves (except grades B and G, see 1.1.6) shall be performed in accordance with the requirements specified in MIL-V-868 when tested under the following conditions:

(a) Non-metering valves and valves without stroke limiting adjustment.

Flow	- Maximum flow (see 2.1).
Supply pressure	- As required to obtain maximum flow.
Return pressure	- 100 lb/in^2 gauge maximum.

(b) Metering valves and valves with stroke limiting adjustment.

Flow	- Maximum flow (see 2.1).
Supply pressure	- As required to obtain maximum flow. Pressure drop shall not exceed 2200 lb/in^2 (1200 lb/in^2 for three way valves).
Return pressure	- 100 lb/in^2 gauge maximum.

2.5 Maintainability. The requirements specified in MIL-V-868 shall apply.

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TABLE VI. Leakage rate.

Spool flow path	Leakage measured	Maximum leakage (mL/min)	Remarks
O, J & K 1/	CA to P CA to R CB to P CB to R	6	
1 & 5	CA to R	6	P & CB Blocked
2 & Z	CA to R	6	P Blocked
3	P to R	6	CA & CB Blocked
4	None		
6 & 7	CA to P	6	CB Blocked
8	CA to R CB to R	6	P Blocked
9	CB to R	6	P & CA Blocked

1/ Leakage across the restricted flow path (orifice) for spool flow paths J and K shall not be less than 8 mL/min nor more than 30 mL/min.

Preparing activity:
Navy - SH
(Project 4810-N076-02)

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